

# COVID-19 CT SCANS SEGMENTATION USING HYBRID DEEP LEARNING OBJECTIVE

## 1. Introduction

The global outbreak of COVID-19 challenged the entire world with not only a health crisis but also a diagnostic dilemma. While RT-PCR remained the gold standard, it often took time and lacked adequate sensitivity in the early stages of infection. This created a demand for complementary diagnostic tools that are both fast and reliable. One such method has been the analysis of lung CT scans. CT imaging provides a detailed view of the internal structures of the lungs, making it an effective method for identifying infection-induced abnormalities.

However, the manual segmentation of these images is laborious, time-consuming, and prone to human error. This project addresses that very challenge—designing a deep learning system that automates the segmentation process, aiding radiologists and medical experts in a rapid, accurate diagnosis.

We present a hybrid deep learning model combining **U-Net** and **ResNet50**, leveraging the strengths of both architectures. The purpose is to produce precise segmentation masks from chest CT images that highlight regions infected by the coronavirus. This combination allows for robust feature extraction (from ResNet) and spatial information preservation (via U-Net's encoder-decoder structure).

## 2. Motivation and Significance

Healthcare professionals faced immense pressure during the peak of COVID-19, often dealing with high volumes of patient data with limited time. As we strive for more resilient and responsive healthcare systems, AI-driven diagnostic tools offer great promise. A model that automatically segments COVID-19 infected areas on CT scans can:

- Reduce the workload on radiologists.
- Enhance the accuracy of diagnosis.
- Provide real-time results in critical environments.
- Assist in monitoring infection progression and treatment outcomes.

This project is not just a technical endeavor but a contribution toward supporting frontline healthcare systems through intelligent automation.

## 3. System Architecture

The system is built on a **hybrid architecture of U-Net and ResNet50**. Here's how each component plays its role:

### **ResNet50 (Feature Extractor - Encoder):**

ResNet50 is a powerful convolutional neural network pre-trained on ImageNet. Its residual connections help it learn deep, complex patterns efficiently. In our model, we use ResNet50 as the backbone to extract meaningful, high-level features from CT images.

### **U-Net (Decoder & Skip Connections):**

U-Net is known for its encoder-decoder structure with skip connections. It helps recover spatial information lost during downsampling and combines it with high-level features for accurate segmentation. We build upon ResNet50's encoder output and pass it through a U-Net style decoder, resulting in highly detailed mask predictions.

#### 4. Dataset Overview

We worked with **two separate datasets**:

- **Dataset 1 (Training Set):**
  - CT scan frames path: /content/drive/MyDrive/COVID/frames\_train/
  - Corresponding masks path: /content/drive/MyDrive/COVID/masks\_train/
- **Dataset 2 (Testing Set):**
  - CT scan frames path: /content/drive/MyDrive/COVID2/frames\_test/
  - Corresponding masks path: /content/drive/MyDrive/COVID2/masks\_test/

These datasets contain axial lung CT images and binary masks representing infected areas. Each image is resized to **128x128** pixels to maintain consistency and optimize model performance.

#### 5. Implementation Details

##### Tech Stack:

- **Languages & Frameworks:** Python, TensorFlow, Keras, OpenCV
- **Visualization:** Matplotlib
- **Interface:** Gradio (for interactive mask generation)
- **Environment:** Google Colab (leveraging free GPU support)

##### Training Parameters:

- **Optimizer:** Adam
- **Loss Function:** Binary Cross-Entropy
- **Batch Size:** 8
- **Epochs:** 6
- **Input Size:** 128x128x3
- **Activation:** Sigmoid (for final layer)

#### 6. Model Performance

After training the model on Dataset 1 and validating on Dataset 2, we achieved:

- **Accuracy:** 98.31%

- **Visual Evaluation:** Predictions closely aligned with ground truth masks, successfully identifying infected lung areas even in complex and low-contrast scenarios.

## 7. Visualization and Interpretability

We implemented a plotting function to visualize:

- Input CT scan
- Ground truth mask
- Predicted mask

This visual assessment was crucial in validating the model's practical usability. Observing the output made it evident how closely the model mimicked human segmentation, reinforcing its applicability in clinical settings.

## 8. User Interface (Interactive Gradio App)

To ensure this tool is accessible and user-friendly, we integrated it with **Gradio**, a Python library that allows us to deploy a GUI in a few lines. Users can upload their own CT scans and instantly see the predicted segmentation.

This interface allows:

- Real-time image upload
- On-the-fly prediction
- Side-by-side comparison

It's built for healthcare professionals who may not have technical knowledge but need actionable results quickly.

## 9. Challenges and Learnings

### Challenges:

- CT images vary greatly in contrast and resolution.
- Some masks had inconsistencies or annotation noise.
- Training a deep network required careful balancing to avoid overfitting.

### Learnings:

- Hybrid architectures offer substantial improvements in medical imaging tasks.
- Visual evaluation is as important as numerical accuracy in medical diagnostics.
- User experience matters — creating interfaces enhances usability beyond the lab.

## 10. Future Scope

Our model lays the foundation for several future enhancements:

- **Add Attention Mechanisms:** To allow the model to focus more effectively on infected regions.
- **Multiclass Segmentation:** Extend beyond binary masks to differentiate between infection stages.
- **Explainable AI Integration:** Provide confidence heatmaps or Grad-CAM visualizations for transparency.
- **Cloud Deployment:** Host the system on a cloud platform to enable remote diagnosis in under-resourced settings.

## 11. Conclusion

This project is a testament to how artificial intelligence can bridge the gap between medical need and technology. The hybrid U-Net + ResNet model proved to be a powerful architecture for medical image segmentation, especially in the context of COVID-19 where rapid diagnosis is vital.

We believe tools like this will become increasingly essential, not just in pandemics, but as a routine part of modern healthcare. By automating labor-intensive tasks, we empower healthcare providers to focus more on patient care, strategy, and intervention.